

A Full Desktop Environment in Go!

(windows, desktops, compositing, ...AI?)

Another Desktop? Why?

- Linux on the Desktop!
- Performance AND productivity
- Developer oriented desktop
- The ultimate proof of **Fyne.io**



And Who is andy.xyz?

- Go developer for 8 years
- Creator of Fyne toolkit
- CTO with multiple exits
- Founder Apptrix.ai
- Maven, Enlightenment
- Fyne.io is my 3rd GUI toolkit



Let's Re-create the Desktop!

Architecture

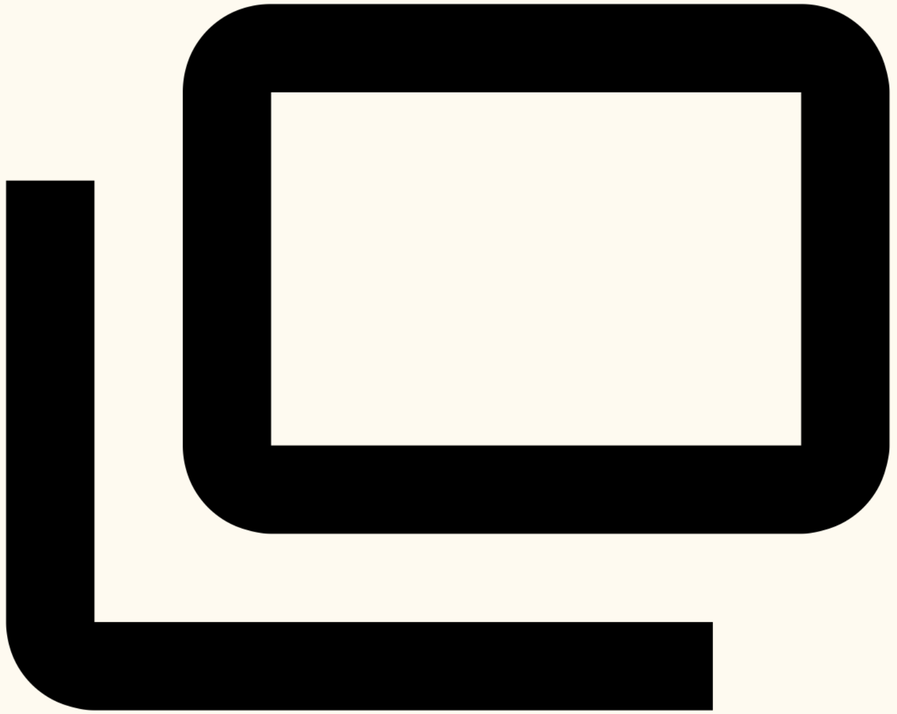
- *Server (X11 / Wayland)*
- Window Manager
- Compositor
- Virtual Desktops
- App icons and Launcher
- Modules & Settings

```
cmd
  tyde
  tyde_runner
  tyde_ctl
internal
  ui
    themes
  icon
  x11
    wm
    composit
  (wayland)
modules
test
wm
```

Window Manager

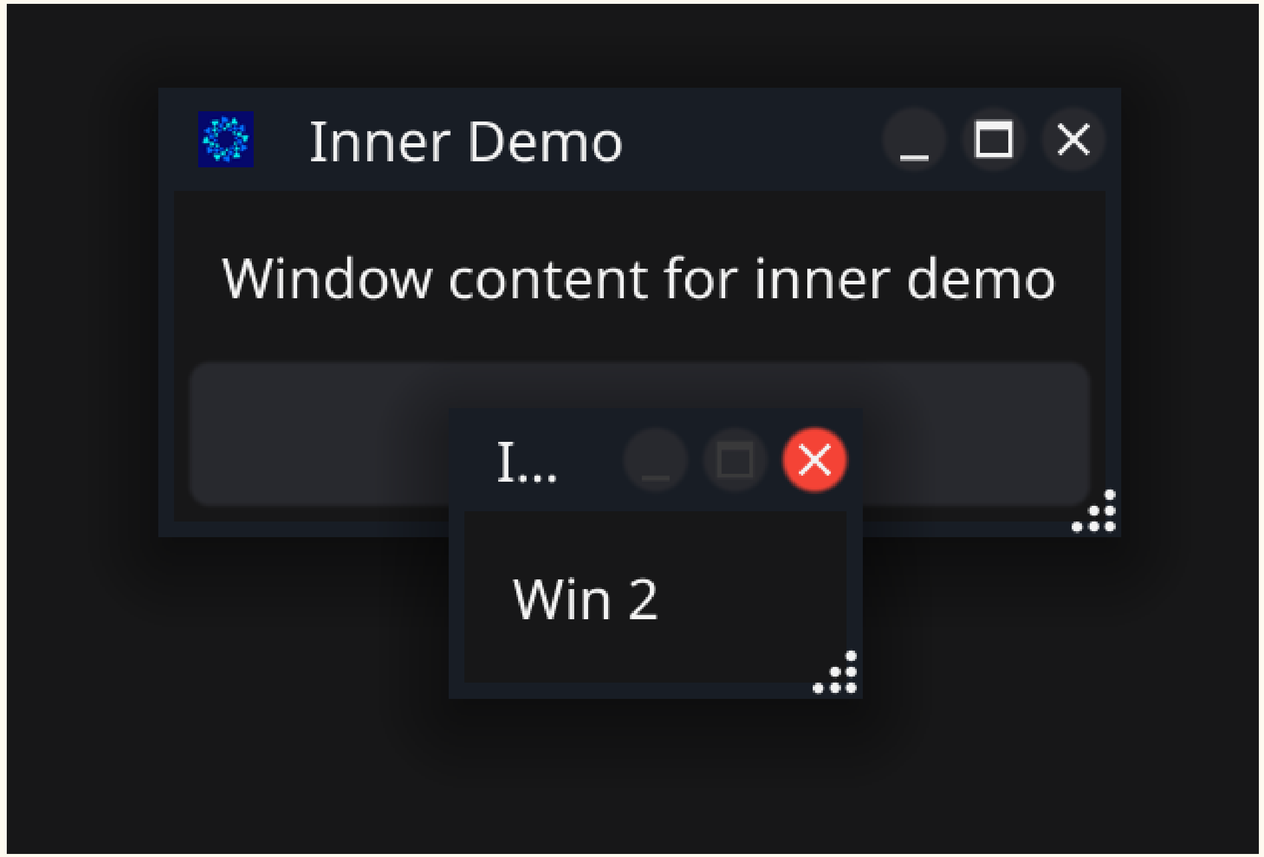
Window Manager Responsibilities

- Positioning and state management
- Drawing borders and menus
- Focus control
- Handling input events
- System integrations



Window Manager (abstraction)

- Render borders
- Handle state
- Stacking and fullscreen etc
- Icons / app launcher
- widget panels



Window Manager (x11)

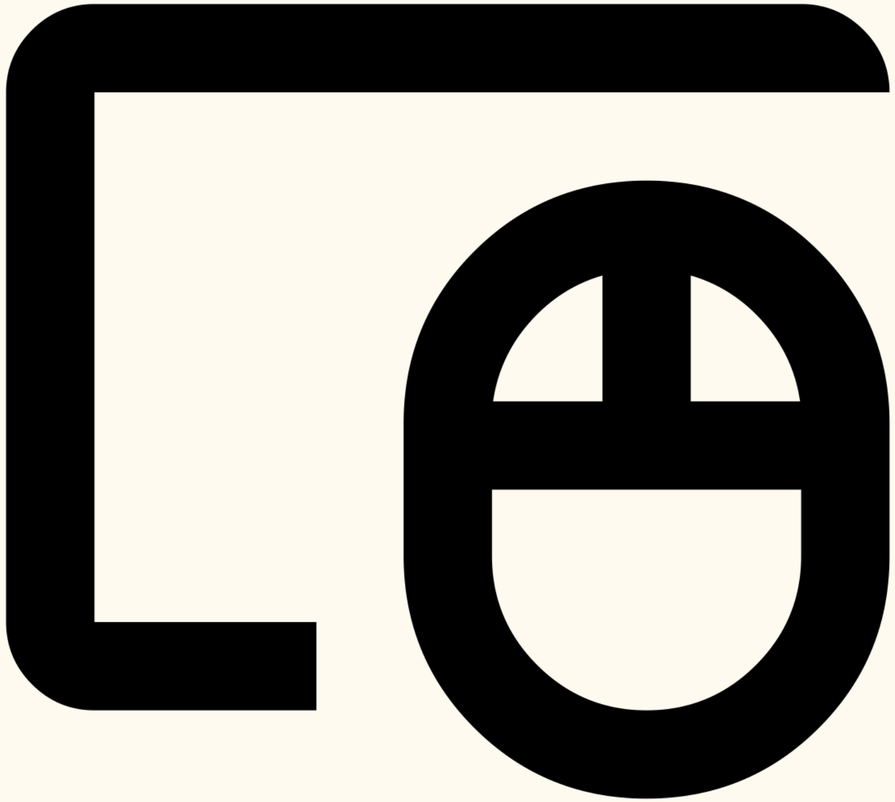
- Event handling
- Translate state to x11 API
- System integration
  - property handling
  - device info
- Optimisations (important!)



Compositor

Compositor: capture

- Capture visible windows
- Event routing (panels etc)
- Handling buffers
- Manage effects



Compositor: draw

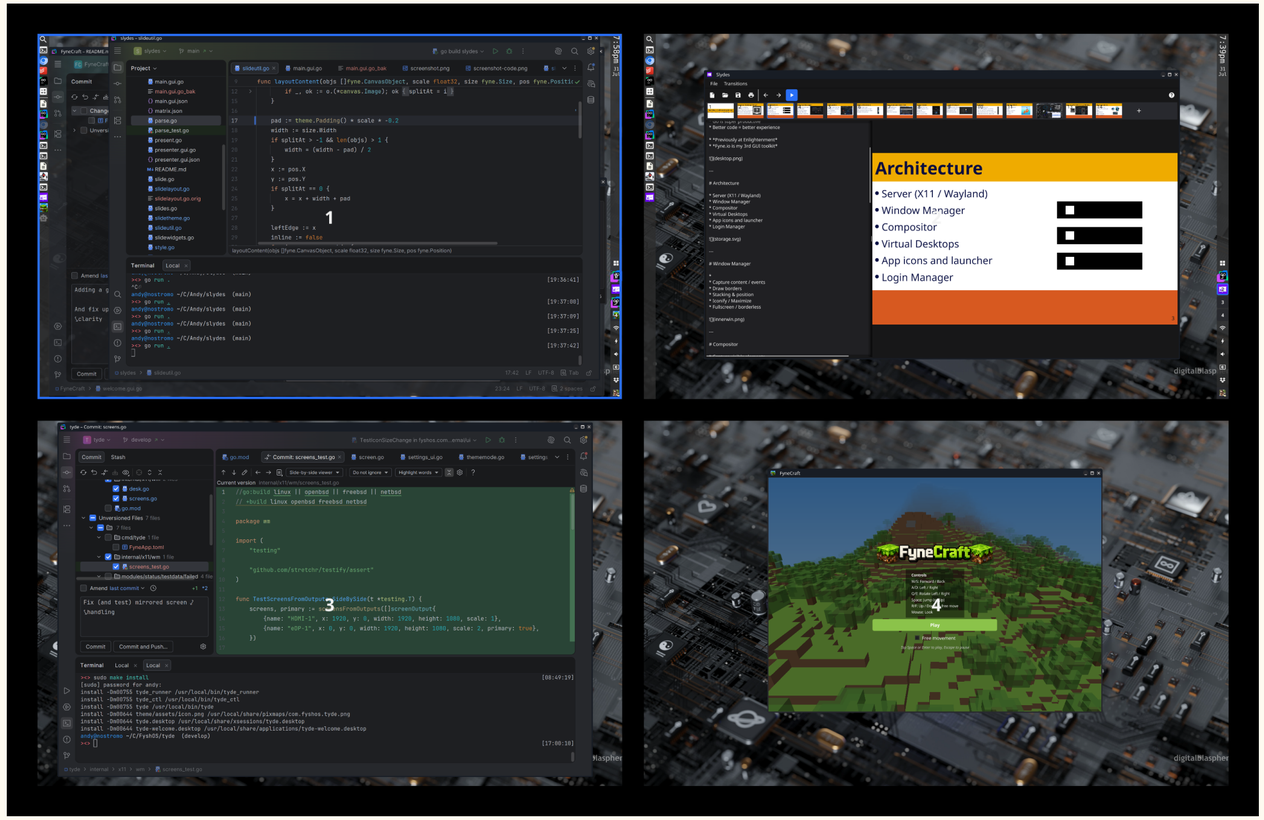
- Background
- Underlays (files, widgets)
- Windows
- Panels & overlays
- Animations



Virtual Desktops

What is a Virtual Desktop?

- Just a big virtual space
- More windows, more pixels!
- Move the image, not window
- *And sync actual X11 windows*



Navigating virtual desktops

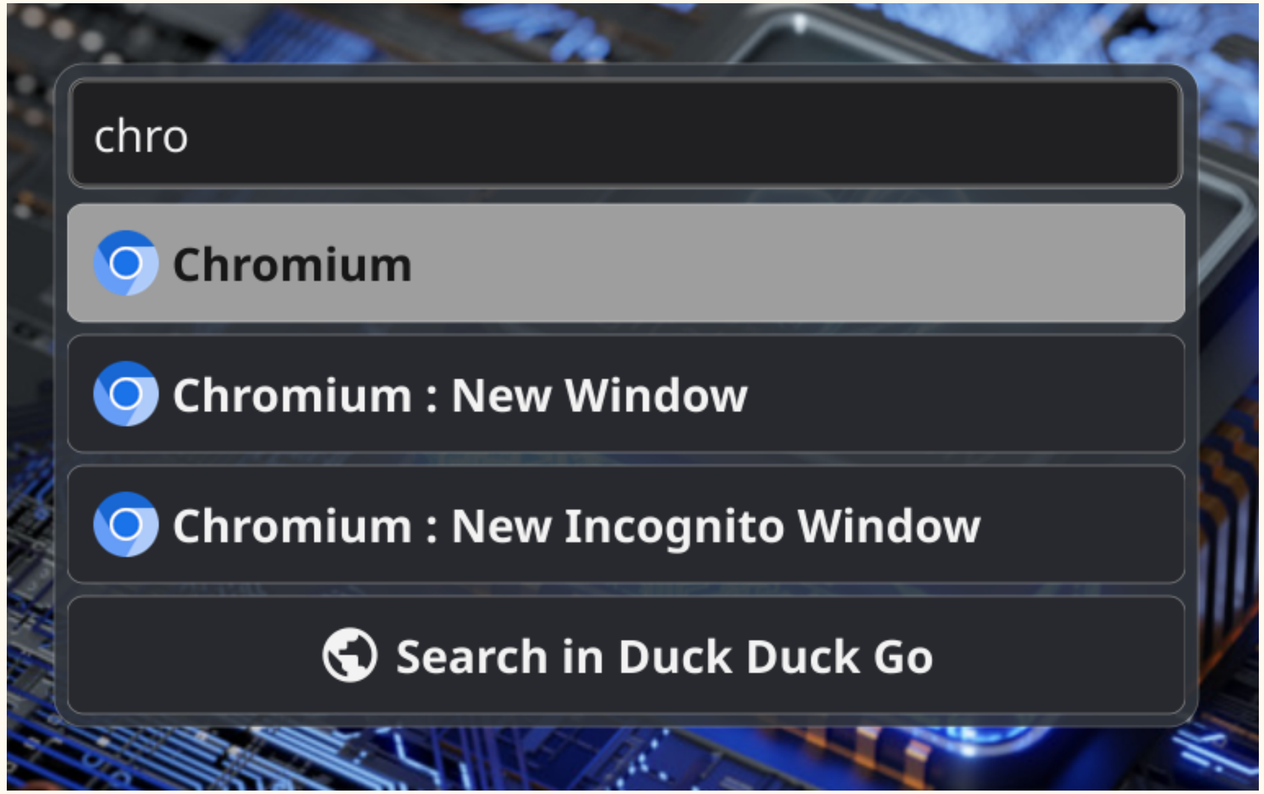
- Arrange in logical space
- Desktop pager previews
- menu items & keys to move
- Key bind to flip & zoom



App icons and launcher

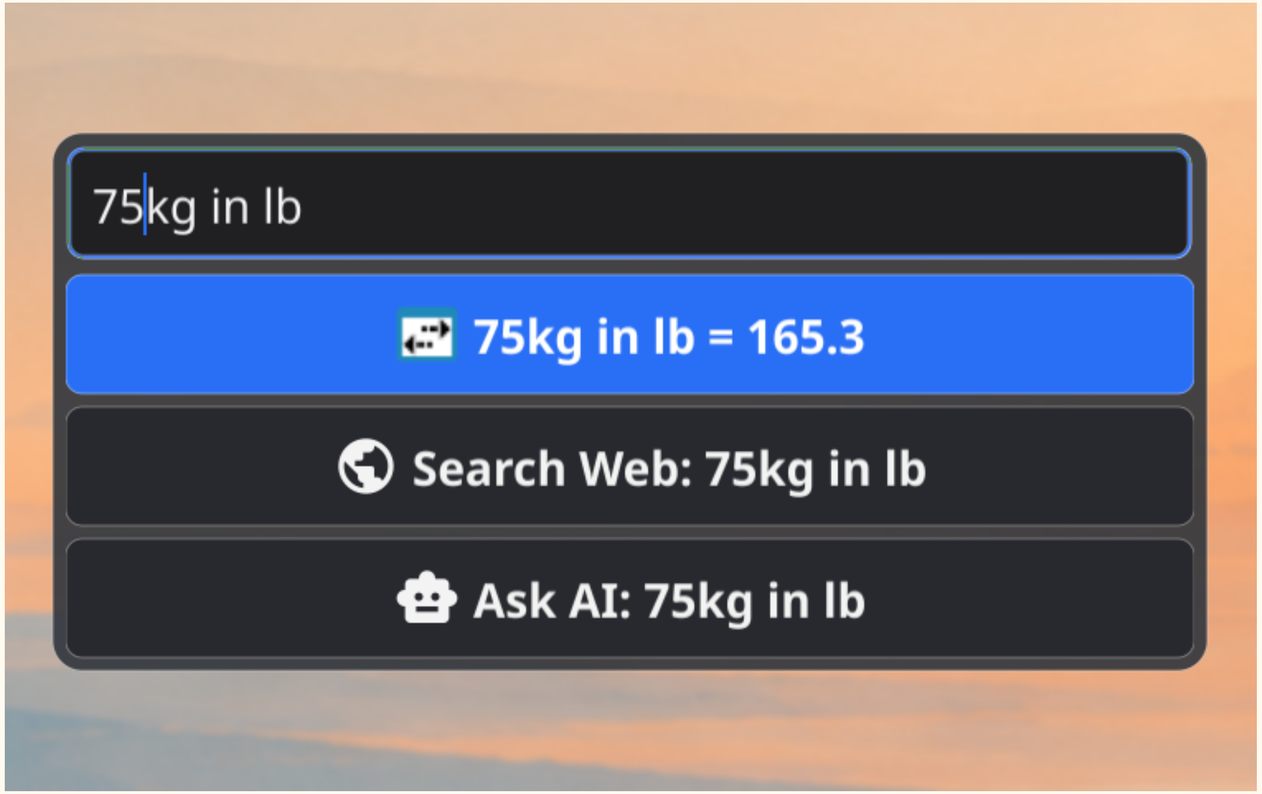
Managing Apps

- Find system and user apps
- Load metadata and icons
- Handle launch configurations
- Provide to app bar
- Launcher suggestions



Suggestions in Launcher

- Conversions (size, price etc)
- Calculations
- Launch URLs
- Show QR codes, big text
- Search the web
- AI suggestions/chat



Launcher extension

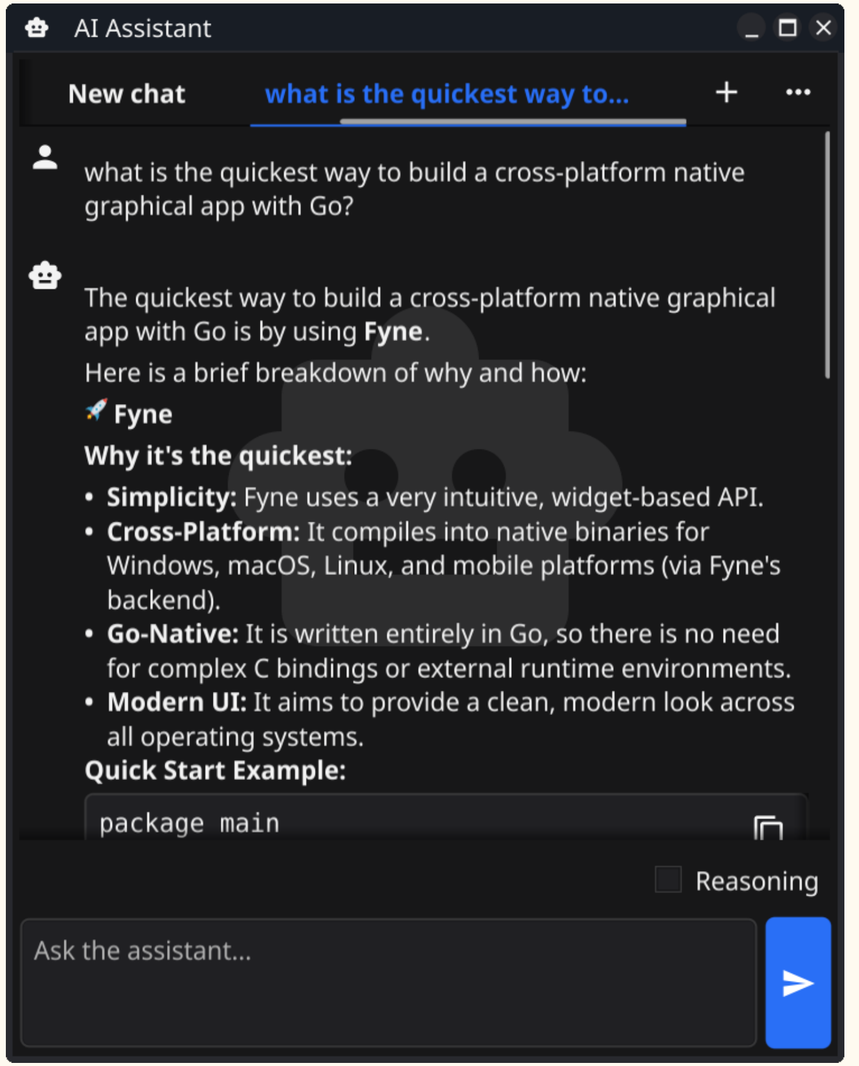
```
type LaunchSuggestion interface {
    Icon() fyne.Resource
    Title() string
    Launch()
}

// module info goes here
```

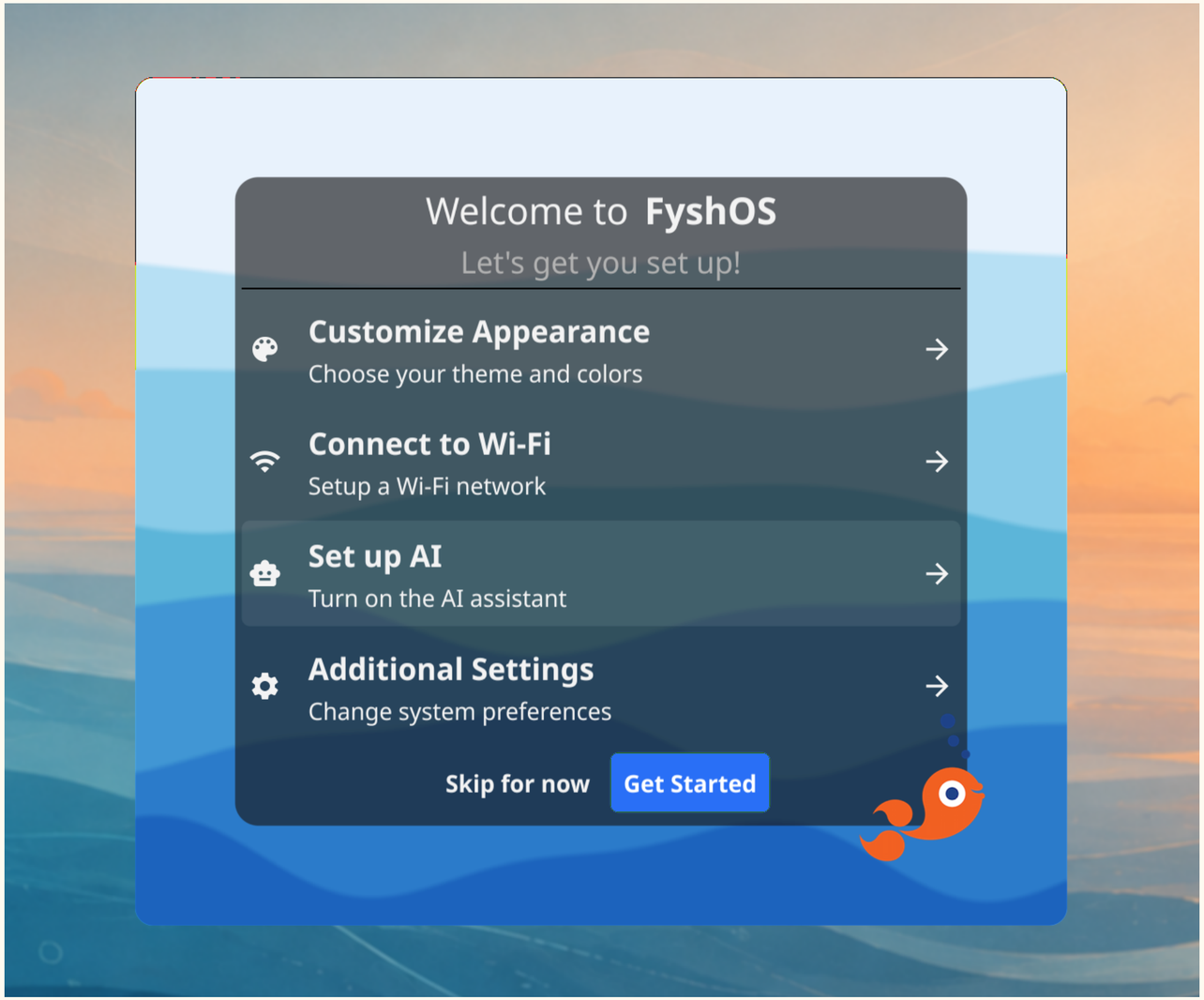
AI Integration

AI Integration - local and frontier

- Local with Kronk or Ollama
- Connect OpenAI or Anthropic
- Fathom: Chat and quick answers
- Choose model and reasoning
- Control the desktop (in progress)



AI Integration - from the start

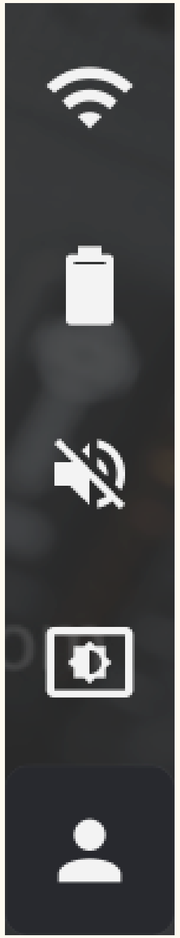


Modules & Overlays

Modules

Main extensibility points:

- Status area (system tray, wifi etc)
- Background / Overlay (fyles, quake term)
- Launcher (conversion, web search, AI)
- Keyboard (brightness, volume etc)
- Accessories (eSheep)



Status area (system tray, wifi)

```
type StatusAreaModule interface {
    Module
    StatusAreaWidget() fyne.CanvasObject
}
```

Screen area (Background, Fyles)

```
type ScreenAreaModule interface {
    Module
    ScreenAreaWidget() fyne.CanvasObject
}
```

Overlays (Quake term)

```
1 type OverlayAreaModule interface {
2     Module
3     OverlayAreaWidget() fyne.CanvasObject
4 }
```

Launcher (conversion, web, AI)

```
type LaunchSuggestion interface {
    Icon() fyne.Resource
    Title() string
    Launch()
}

type LaunchSuggestionModule interface {
    Module
    LaunchSuggestions(string) []LaunchSuggestion
}
```

Keyboard shortcuts (volume etc)

```
type KeyBindModule interface {
    Shortcuts() map[*Shortcut]func()
}
```

Window Accessories (eSheep)

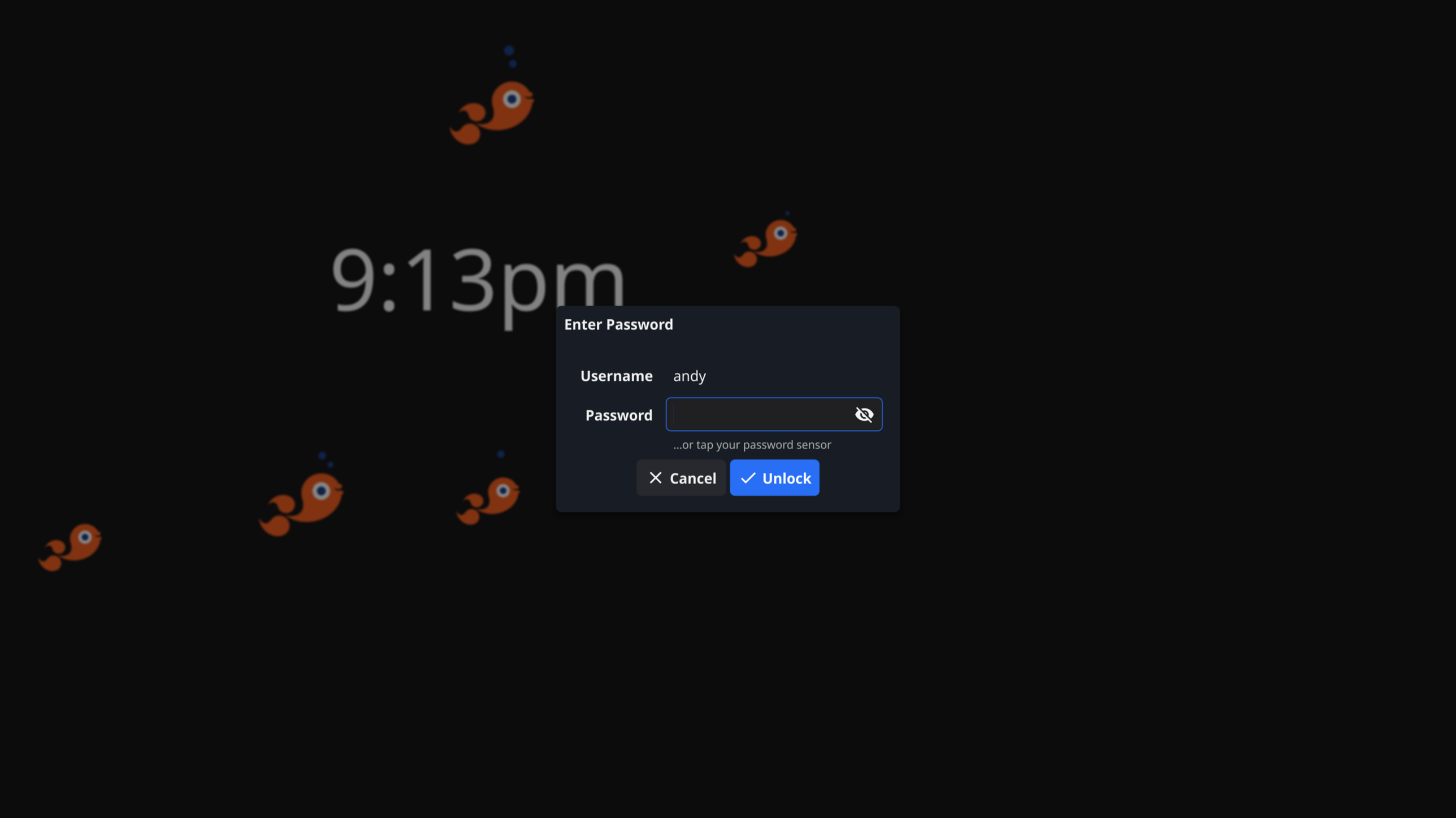
```
1 type WindowAccessory struct {
2     Object fyne.CanvasObject
3     Window Window
4 }
5
6 type WindowAccessoryModule interface {
7     Module
8     WindowAccessories() []WindowAccessory
9 }
```


Fyne Terminal: ~
Welcome to fish, the friendly interactive shell
Type help for instructions on how to use fish
andy@nostromo ~
><>
[18:01:35]

ScreenSaver

ScreenSaver

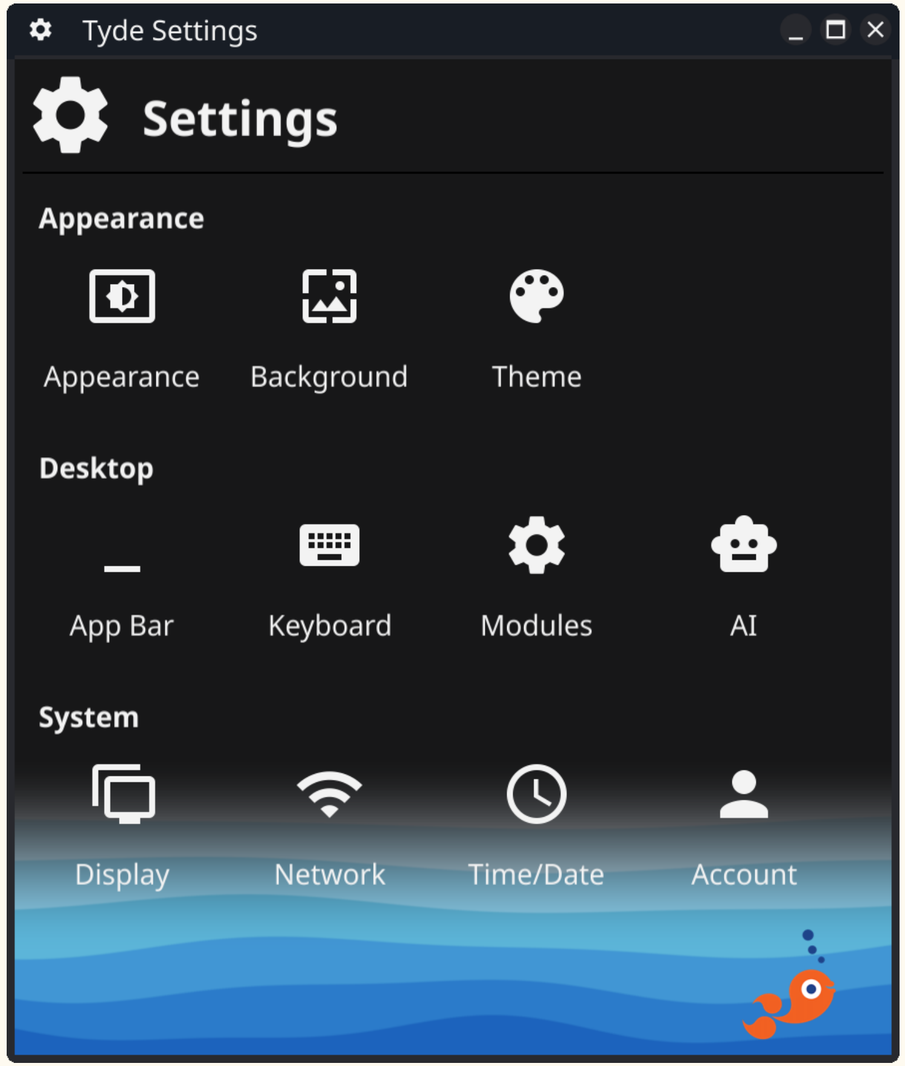
- Just another Fyne app
- One window per screen
- Fullscreen - swallow all events
- Password and fingerprint support
- Drop-in custom visual


9:13pm
Enter Password
Username
andy
Password
...or tap your password sensor
Cancel
Unlock


›_
andy.xyz

AND System Settings

- Appearance
- Desktop behaviour
- User management
- Hardware management
- System settings

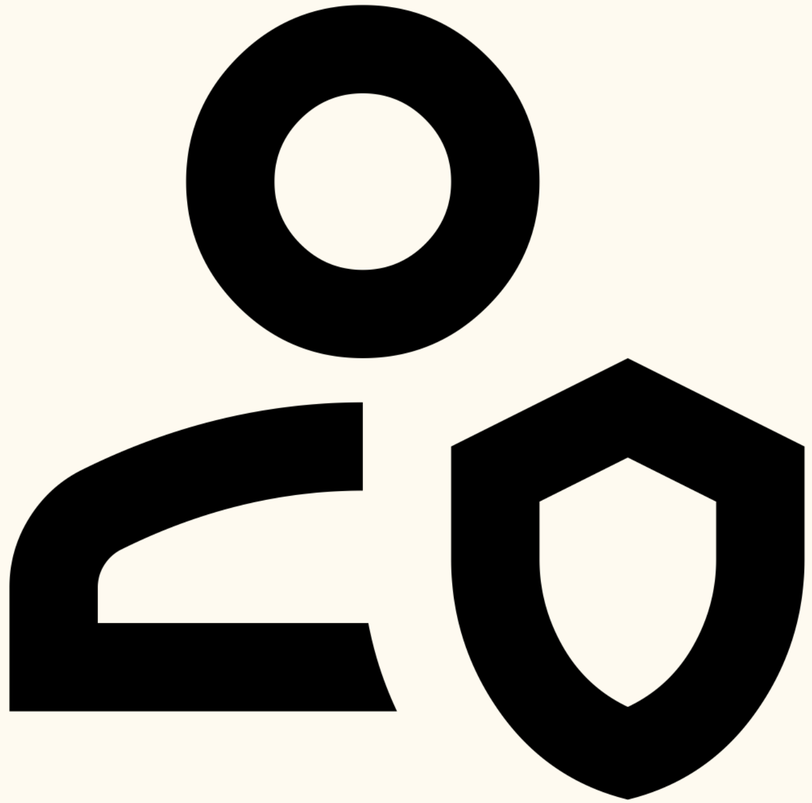


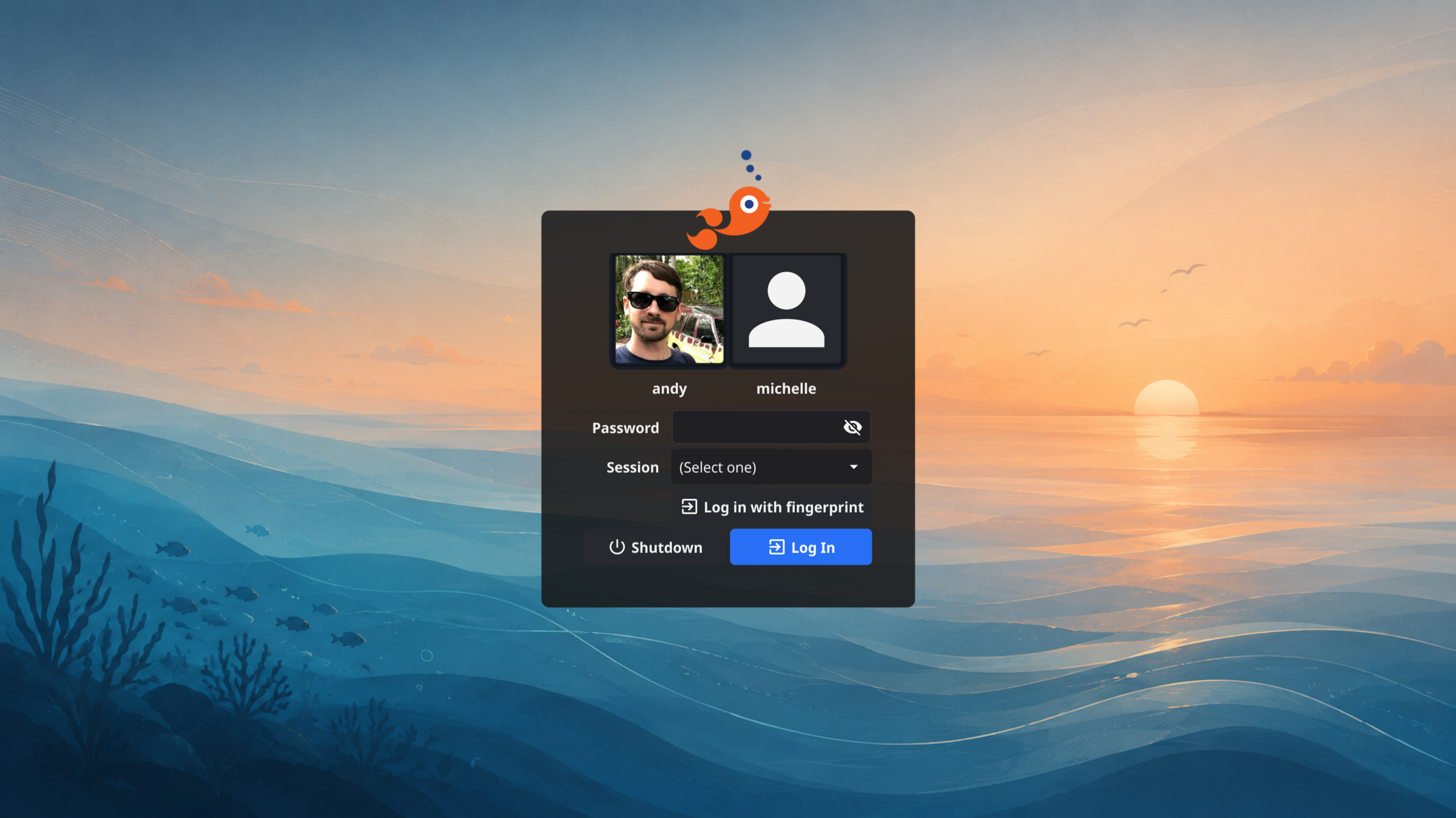
Login Manager

Login Manager Responsibilities

- Launch graphical context
- Display users and options
- Handle authentication

- Initiate user session
- Actually launch desktop




andy
michelle
Password
Session
(Select one)
Log in with fingerprint
Shutdown
Log In

All Together = FyshOS

FyshOS repositories

- tyde & fin
- backgrounds, appie
- dryvers, screens
- fancyfs, fyles
- fyshsaver
- ferry



Home

Trash

Fyne Terminal: ~

```
Welcome to fish, the friendly interactive shell
Type help for instructions on how to use fish
andy@nostromo ~
><>                                          [08:09:28]
```

Fyles : Dropbox

/home/andy/Dropbox

Favourites

Home

/

(Parent)

050-053_ Snapsh...

2024-02-05 21-5...

264320_773080

2IK8ZF060683a5...

AI ML Workshop

Apps

Articles of Associ...

Tyde Settings

Appearance

Background

Display

Theme

App Bar

Keyboard

Modules

Themes

Default

Description...

Install

Matrix

Description...

Install

Neon

Description...

Install

More themes...

Customise...

Slydes

File

```
+++
theme = "sgtheme"
+++

# Building a Full Desktop in Go!
## (windows, desktops, compositing,

---

# Another Desktop? Why?

* Existing ones are old
* Go is super productive
* Better code = better experience
```

Building a Full Desktop in Go!

(windows, desktops, compositing, modules)

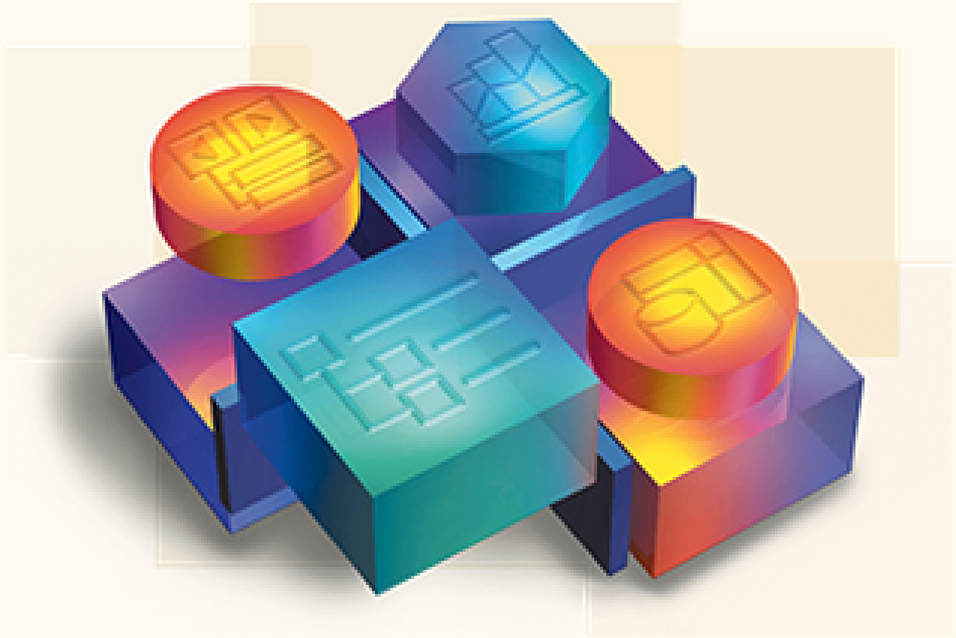
8:10am

20 May

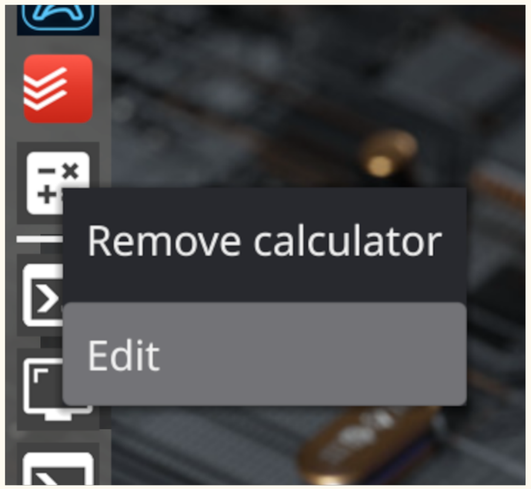
Bonus: WYSIWYG App Editing

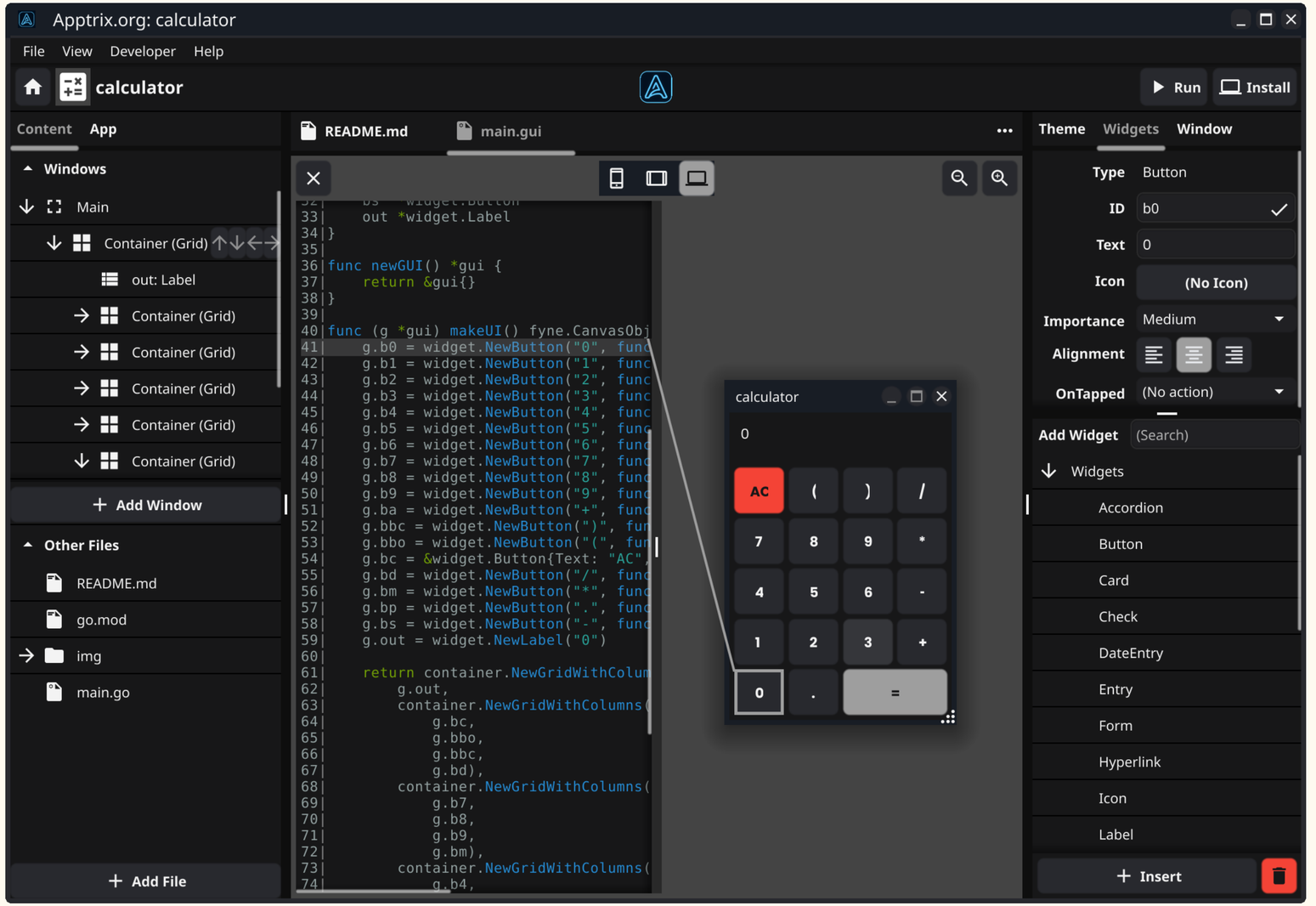
WYSIWYG App Editing

- Visual app editing
- Drag and drop
- Preview for desktop & mobile
- Compile to native apps
- Works from FyshOS source code



App Editing - Apptrix.org





Find out more

- https://fyshos.com
- github.com/FyshOS/tyde
- #fyne
- @andydotxyz
- https://andy.xyz
- https://apptrix.ai







